Homework: Time-frequency transform (1) – Using Kaiser window for spectral analysis

A text file HW1_Kaiser.txt is provided to you, which consists of a timing waveform and your task is to analyze its spectrum using the Kaiser window. Specifically, the signal is known to be in the form of

$$x(t) = \sum_{i=1}^{Q} A_i \sin(2\pi f_i t)$$

Your task is to estimate Q, A_i, and f_i. In your report, provide the following:

1. A plot of the time domain signal.
2. At least four figures of spectrum plot using different parameters (α) of the Kaiser window. In each plot, label the x-axis, y-axis, and value of α clearly. Adjust the xy axes limits properly.
3. The estimated Q, A_i, and f_i.
4. Your Python source code.

Notes:

1. The specific procedure is:
   - Read the file into Python
   - Multiply the waveform by the Kaiser window
   - Perform spectral analysis on the windowed signal
2. The N-point discrete Kaiser window is defined by

$$w(n) = \frac{I_0\left(\pi\alpha\sqrt{1-\left(\frac{2n}{N-1}-1\right)^2}\right)}{I_0(\pi\alpha)}, n = 0, 1, \ldots, (N-1)$$

   where $I_0(\cdot)$ is the zeroth-order modified Bessel function of the first kind, and α is a parameter of adjustment. Implement the above equation directly. You can use the built-in function `scipy.special.i0` but you shall not use the built-in Kaiser window directly. See Wikipedia for details about the Kaiser window.
3. Remember to compensate the window gain, which depends on the parameter α used.
4. In the class note, we introduce the procedure to estimate the continuous Fourier transform $X(f)$. For this homework, since we are going to estimate A_i ultimately, we need to convert $X(f)$ to A_i. Note that for each frequency component, the truncated continuous FT is

$$\int_0^T A_i e^{j\omega_i t} e^{-j\omega t} dt$$

   When evaluated right on ω_i, i.e. $\omega = \omega_i$, the result is

$$A_i \int_0^T dt = A_i T$$

   Therefore, to convert the estimated continuous FT to the amplitude A_i, we simply divide the result by the total time duration T.
5. A Python sample code is provided to you. You don't necessarily have to use it.